



Star Factory

A number of newborn stars ten times the size of the sun have been spotted in this dust cloud known as the Rosetta Nebula.

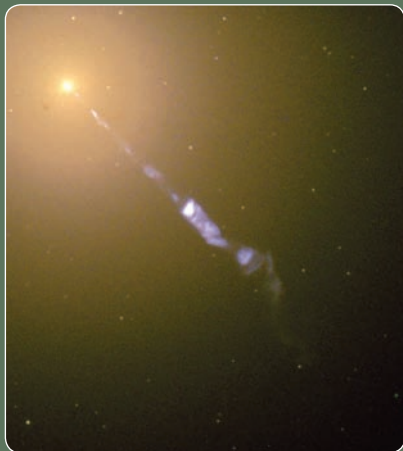


Nokia Comes with Music

The Comes with Music service launched in China allows users to download thousands of DRM free music for free. The service is going to be launched in India soon.

Bleeding Edge

Fresh theories, concepts and speculations from the world of science



THE OMG PARTICLE: Theoretical quantum physicists are having some problems because their theories are falling short of observed fact. The oh my God particle is a form of radiation with so much energy, that there is no way that scientists can explain it. These particles are observed in gamma ray showers, or to be very accurate, extreme-energy cosmic rays. Apart from being unable to explain how the particles have such high energies, the scientists are at a loss to explain where the particles come from. Although not proven, the most likely candidate for the origin of these particles are supermassive blackholes in the center of galaxies. Scientists have almost no proof for

The gamma ray emission jet from the active galactic nuclei of the M87 galaxy. If you travel along the jet at the speed of light, it will take you five millennia to traverse.

these claims though, and it is equally probable that there is a Ultra-high-energy cosmic ray gun at every galactic center. The scientists are good at covering up their lack of knowledge with a sense of humour though, and the OMG particle is named as a comic retaliation to the God particle. The LHC at CERN made the search for the God particle famous. The atom smashing scientists are breaking open atoms to understand how they work, which is similar to throwing a computer from a building, looking at all the components, and figuring out how the computer works. This approach might not make too much sense, but that is what they are doing.

THE SEARCH FOR EXTRATERRESTRIAL LIFE:

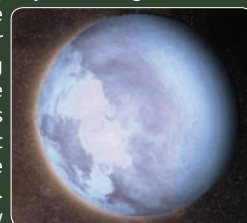
This is not to be confused with the search for extraterrestrial intelligence. The SETI program was dedicated to analyzing radio waves from space in the hope that if an ET was calling, we would at least pick up the signal. However, we ourselves used radio waves for broadcasting for a very short period, and mankind switched to fibre optic and other undersea cables as the principle mode of communication. Scientists are aggressively looking for extraterrestrial life,



A famous photograph of a Martian meteorite, which possibly shows the signs of bacterial life.

both intelligent and otherwise in a variety of ways. Bacteria is a good place to start, as these little creatures live in the most demanding geographic locations, lava vents that are deep underwater, cracks in the Antarctic ice cap, and in the atmosphere close to the boundary of space. There are clouds of amino acids and alcohol (enough to make the all the people who will ever live drunk for life) floating around in space - especially in nebulae, and this is similar in composition to what scientists call "organic soup", the primordial base out of which life is believed to have emerged. Bacteria could even be found in our solar system, in a comet or wherever ice is, on the Moon, Mars or Europa. The new planetary rovers have small labs in the robotic arms that can analyse soil samples for signs of life, using recently developed tests and techniques. Meanwhile, the hunt for extrasolar planets (planets around other stars) has become extremely aggressive, with monthly updates to an ever-growing catalogue. The problem is in locating tiny planets that appear very close to the large stars. To understand more about this, view the Astronomy 160 lectures in our DVDs.

The 2006 COROT mission launched by the French Space Agency is marking down small planets. There are two major planet hunting missions in the offing, NASA's Terrestrial Planet Finder and The Kepler Mission.



Want to know the currently identified extra-solar planet most likely to contain life? It's called Gliese 581 d. Gliese 581 is a Red Dwarf planet in the Libra constellation with a number of planets orbiting it.

According to a small group of scientists though, life in the universe is superabundant, and we could always see it. Specks of dust in space are suspiciously close in size to bacteria. These scientists believe that space dust is just dormant bacteria, and a comet or a meteor smashing into any planet (even the extrasolar ones) seeds life on that planet.

DARK FLOW: Astronomers have a problem, galaxies are not moving in the sky the way they should. There is a mysterious object known as the "great attractor" which is termed a "gravity anomaly" as it is the origin of a large amount of gravitational force. Unfortunately, this gravitational anomaly is hiding behind the Milky Way, so we cannot really observe it too well as our own galaxy obscures it. Dark Flow is a related concept, only much larger than the great attractor. There is some mysterious force, at the edge of the visible universe, that is sending entire galaxies hurtling along at

nearly 3.2 million kilometers per hour. The great attractor has a known influence of at least fifty thousand Milky Ways, but the Dark Flow anomaly is greater. Dark Flow is gravitational influence from outside the observable universe, that is changing our skies - at least twenty degrees of it. In fact, the scale and the distance of the distortion suggests that this may be influence from material outside the universe. Some scientists even believe that we may be seeing two universes rubbing shoulders here. The universe is maintaining it's mysteries despite the dissection by the scientists.



Centaurus A is close to the region in the sky where the Dark Flow was detected